

Biological control of plant diseases – What has been achieved and what is the direction?

David B. Collinge¹  | Dan Funck Jensen²  | Mojgan Rabiey³  | Sabrina Sarrocco⁴  | Michael W. Shaw⁵  | Richard H. Shaw⁶ 

¹Department of Plant & Environmental Sciences and Copenhagen Plant Science Centre, University of Copenhagen, Frederiksberg, Denmark

²Department of Forest Mycology and Plant Pathology, SLU, Uppsala, Sweden

³School of Biosciences, University of Birmingham, Birmingham, UK

⁴Department of Agriculture, Food and Environment, University of Pisa, Pisa, Italy

⁵School of Agriculture, Policy and Development, University of Reading, Reading, UK

⁶CABI, Egham, UK

Correspondence

David B. Collinge, Department of Plant & Environmental Sciences, University of Copenhagen, Thorvaldsensvej 40, 1871 Frederiksberg C, Denmark.
Email: dbc@plen.ku.dk

Abstract

The global sustainability agenda is increasing the demand for reduction in inputs into agricultural production while maintaining profitable yield of quality products. Plant diseases are a major constraint for both yield and product quality, but often tools for their control are ineffective or lacking. Biological control using antagonistic micro-organisms has long been a subject of research resulting in a wide range of products that are now available and marketed in specific territories around the world. These preparations are often niche products with narrow uses. The research effort is intense both to develop new biological control agents (BCAs) and to obtain knowledge of the mechanisms underlying biological disease control. The prospects for biological control are promising. As a minimum, BCAs supplement other sustainable disease management practices such as disease resistance, and present opportunities for controlling diseases for which other approaches are ineffective or unavailable. We can realistically expect increasing use of BCAs to control crop diseases in ways that will benefit the environment. This review paper arose from a webinar held by the British Society for Plant Pathology as part of the International Year of Plant Health (IYPH2020), at which many of the 300 participants posed or discussed interesting questions. This review is based on that input and the panel members at the webinar are all included as co-authors in this review.

KEYWORDS

plant diseases, plant pathology, sustainable disease control, virus

1 | INTRODUCTION

The green agenda, specifically the need to focus on sustainable use of the resources available on our planet, is receiving increasing attention. The discipline of plant pathology can contribute to this agenda by improving agricultural efficiency, both in terms of increased yield and reduced environmental impact, in particular by reducing the estimated 20%–30% losses caused by pests and diseases (Oerke, 2006; Savary et al., 2019) and the side-effects of disease and pest control actions. Both can be achieved by reducing inputs per unit of production (e.g., watering, spraying pesticides and applying

inorganic fertilizers) and reducing food and fodder spoilage after harvest. Disease resistance is also an important means of disease management but effective resistance is often not available, whether introduced by conventional means (plant breeding) or biotechnologically by genetic engineering including new genomic technologies (Collinge & Sarrocco, 2022).

Biological control is receiving increasing attention as an alternative means of disease control, both pre- and postharvest, especially where disease resistance or chemical control are not available. This review was motivated by a webinar held on 21 September 2020 as part of the British Society for Plant Pathology's (BSPP) contribution

to plant health week and the UN initiative "International Year of Plant Health 2020" (IYPH2020). The authors were on the panel and were inspired by enthusiastic participants from around the globe – see BSPP News #93 (2021). The recording is available via <https://www.bspp.org.uk/conferences/webinar-biocontrol/>. Many interesting issues were brought up by the participants, who represented undergraduate and graduate students, researchers, practitioners and industry as well as others challenged by or fascinated with plant diseases. We discuss many of the points they raised.

2 | WHAT IS BIOLOGICAL CONTROL?

For plant diseases, biological control is most usually defined as direct or indirect inhibition of a disease, or the pathogen causing the disease, by another organism (antagonist) or group of organisms (Cook & Baker, 1983). The beneficial organism is termed the biological control agent (BCA; Jensen et al., 2016; Tronsmo et al., 2020). A broader definition also includes specialized metabolites, isolated, for example, from interactions or plant extracts that can be useful for controlling diseases. These include substances with signalling, antibiotic or attractant activities (e.g., pheromones), and are often termed biopesticides (Roberts & Taylor, 2016). However, we recommend that the misleading term biopesticides is avoided and the new term bioprotectants is used, as proposed by Stenberg et al. (2021). Thus, the term bioprotection should replace the wider use of the term biological control mentioned above, and include the use of non-living extracts and natural products as well as the more narrowly defined biological control; thus, the term biological control can be reserved for situations where a living BCA is applied (Stenberg et al., 2021).

Classical BCAs are defined as natural enemies that self-propagate and establish in the environment to which they are introduced to suppress pest populations. Augmentative BCAs are not expected to establish and are defined as mass-produced natural enemies that are periodically introduced into a specific environment to suppress pest—and pathogen—populations. Augmentative BCAs can be further subdivided into seasonal inoculative agents, which can reproduce and persist throughout the growing season, and inundative agents, which cannot reproduce and must be reapplied frequently throughout the growing season (Stenberg et al., 2021).

Biological control is seen to offer several opportunities for improved disease control, especially where conventional approaches are limited or compromised. Alongside the use of disease-resistant cultivars, biological control is seen to have an important role in integrated pest management strategies aimed at reducing the use of chemical pesticides. A BCA is an organism or collection of organisms rather than a chemical per se. It is likely to be more specific in effect than most commercialized agrochemicals and less likely to leave potentially harmful residues in the environment. A living organism may be able to penetrate the diseased plant or affect the target pathogen in a way that a chemical cannot. In addition, in some situations, the risk of the evolution of pathogens resistant to a chemical pesticide is greatly reduced by applying a BCA. Biological control is also publicly

perceived as natural and therefore less environmentally harmful than chemical control; in many cases this is true, because no completely novel molecule is being introduced to the environment. Because of these favourable perceptions, many forms of biological control are also accepted for use in organic cultivation. It is also claimed that—again only in some cases—a BCA may be cheaper than a pesticide.

3 | HISTORY AND ORIGIN OF BCAS

From 1932 onwards, Weindling published several papers (Weindling, 1932, 1934, 1941) demonstrating that a *Trichoderma* isolate was able to reduce damage to citrus seedlings caused by *Rhizoctonia solani* and describing some of the possible mechanisms of action. Today, *Trichoderma* spp. are probably the most widely used organisms as BCAs for plant disease control worldwide (see below and Table 1) (Lorito et al., 2010). In another example, inoculation of freshly cut pine tree stumps with the commercially available *Phlebiopsis gigantea* has been used as a biocontrol against *Heterobasidion annosum* in pine plantations in parts of Europe since the 1960s (Pratt, 1999), following research by Rishbeth (1963). These and other seminal projects—for example, influential work on take-all of wheat from the 1970s and 1980s (Cook, 2007), and, from the 1970s, biological control of crown gall in stone fruit trees caused by *Agrobacterium tumefaciens* with the BCA *Agrobacterium radiobacter* K84 (syn. *Rhizobium rhizogenes*) (Kerr, 2016), paved the way for a large body of research aimed at demonstrating that beneficial microorganisms could be used to control plant pathogens. During the 1980s, biological control was seen not only as a strategy, but also as a philosophy to reduce crop loss due to plant diseases. In 1981, Papavizas highlighted that biological control could find its roots in earlier farming practices including rotation of crops, burial of infected crop residues and fertilization with organic manures, all allowing time and opportunity for biological destruction of pathogens (Papavizas, 1981). However, in 1974, Baker and Cook had already introduced the term "pathogen-suppressive soils" to describe examples of natural, apparently biological, control of soilborne plant pathogens where a precise mechanism of control was still uncertain (Baker & Cook, 1974). These suppressive soils were initially recognized because of the absence of a disease, despite an environment apparently favourable for its occurrence as well as the presence of a susceptible host and virulent pathogens. Suppressiveness to specific pathogens was explained as the result of a natural "microbiological makeup" of the soil, or of management practices encouraging antagonists, which can control disease (Papavizas, 1981). Over the last 30–40 years, key contributions to understanding the biology of disease or pathogen-suppressive soils include those from pioneering researchers like Claude Alabouvette, Dijon, France and from Washington State, USA, David Weller and Linda Thomashaw together with R. J. Cook cited above. A further step towards transforming interesting research results into tools available for farmers was the appearance on the market of additional crop protection products based on microorganisms. BCA products based on *A. radiobacter* and *P. gigantea* were mentioned

TABLE 1 Examples of commercial biological control products for controlling plant disease

Bioactive ingredient(s)	Target (disease or pathogen)	Mechanism(s) and other information	Territories approved/marketed	Product name (supplier)	Reference
Bacteria					
<i>Agrobacterium radiobacter</i>	Crown gall caused by <i>Agrobacterium tumefaciens</i>	Antibiosis and competition in wounds	Australia 1988, USA 2000, Turkey 2005	K84 or K1026 Galltrol, NOGALL® (Becker Underwood)	Kerr and Bullard (2020)
<i>Bacillus amyloliquefaciens</i> (formerly <i>B. subtilis</i>) QST 713	Many, e.g., <i>Pythium</i> , clubroot; yellow rust; bacteria	Antibiosis (lipopeptides), induced resistance	Global c.2005	Serenade (Bayer Crop Science) ^a	Lahlali et al. (2011), Reiss and Jørgensen (2017)
<i>Bacillus subtilis</i> GB03	Cotton wilts caused by <i>Rhizoctonia</i> and <i>Fusarium</i>	Antibiosis and competition	USA mid-1990s	Kodiak® (Gustafson, USA)	Brannen and Kenney (1997), Miliaković et al. (2020)
<i>Pseudomonas chlororaphis</i> MA342	Many, e.g., <i>Fusarium</i> crown rot	Endophyte in embryo: antibiosis	EU; USA 2001	Cedomon® (Lantmännen BioAgri, Sweden)	Chin-A-Woeng et al. (2003)
<i>Pseudomonas</i> sp. DSMZ13134	Soilborne pathogens	Competition for space and nutrients, induced resistance	EU 2013	Proradix® SP (Sourcon Padena, Germany)	Anastassiadou et al. (2020)
<i>Streptomyces griseoviridis</i>	Many, includes: bacteria, fungi and oomycetes	Antibiosis and competition	Global Finland 1982, USA 1993	Mycostop® (Verdera)	Lahdenperä et al. (1991)
Fungi and oomycetes					
<i>Ampelomyces quisqualis</i> M10	Powdery mildew	Mycoparasitism	Global 1994	AQ10 (CBC Europe)	Sztejnberg (1993)
<i>Aspergillus flavus</i> NRRL 21882	Mycotoxigenic <i>Aspergillus</i> spp. on maize	Competition for nutrients and space	USA	Afla-Guard® GR (Syngenta)	Dorner and Lamb (2006)
<i>A. flavus</i> AF36	<i>Aspergillus flavus</i> on cotton	Competition for nutrients and space	USA 2003	Afla-Guard® (Cicleone Global)	Junaide et al. (2013)
<i>A. flavus</i> MUCL 54911	Mycotoxigenic <i>Aspergillus</i> spp. on maize	Competition for nutrients and space	Italy	AF-X1 (Pioneer Hi-Breed Italia)	Mauro et al. (2018)
<i>Aureobasidium pullulans</i> DSM 14940 + DSM 14941	Fire blight and postharvest diseases of pome fruits	Competition for space and nutrients, physical barrier against pathogen infection	EU	Blossom Protect (Manica)	Kunz (2004)
<i>Candida oleophila</i> I-182	<i>Botrytis</i> spp., <i>Penicillium</i> spp. on citrus, pome fruit	Induced resistance	USA 2001	Aspire (Ecogen, Inc.)	Droby et al. (2002), Gardener and Fravel (2002)
<i>Coniothyrium minitans</i> CON/M/91-08	<i>Sclerotinia sclerotiorum</i> , <i>Sclerotinia minor</i>	Mycoparasitism of sclerotia	Global 2001	Contans® WG (Bayer)	Whipps et al. (2008)

(Continues)

TABLE 1 (Continued)

Bioactive ingredient(s)	Target (disease or pathogen)	Mechanism(s) and other information	Territories approved/marketed	Product name (supplier)	Reference
<i>Glocladium catenulatum</i> J1446 (current name <i>Clonostachys rosea</i>)	Soilborne pathogens and grey mould	Competition in rhizosphere, mycoparasitism, CWDE, antibiosis	EU 1998	Glomix [®] Prestop (Verdera)	Mcquillen et al. (2001)
<i>Glocladium virens</i> GL-21	<i>Rhizoctonia solani</i> and <i>Pythium</i> spp. on ornamentals, vegetables, cotton		USA 1990	GlioGard [™] , Soilgard (Thermo Trilogly Corp.)	Gardener and Fravel (2002), Junaid et al. (2013)
<i>Phlebiopsis gigantea</i>	Root and butt rot caused by <i>Heterobasidion annosum</i>	Competition	EU 1994	Rotstop (Verdera)	Pratt et al. (2000), Żółciak et al. (2020)
<i>Pseudozyma flocculosa</i>	Powdery mildew on wheat, barley, grapevines, apple and vegetables	Parasitism	USA c.2000	Sporodex (Ecogen, Inc.)	Kiss (2003), Laur et al. (2017)
<i>Pythium oligandrum</i> M1	Grey mould and <i>Sclerotinia</i>	Mycoparasitism, induced resistance	EU c.2001	Polyversum [®] (Gowan), Polygandrum (Plant Production Institute, Slovakia)	Brozova (2002)
<i>Trichoderma afroharzianum</i> CBS 134709 (IBT 41409, G.J.S. 08-137)	Soilborne fungal plant pathogens (mostly food crops)	n/a	EU	Canna [®] (Canna International BV NL-Breda)	Degenkolb et al. (2015)
<i>Trichoderma asperellum</i> ICC012 + <i>Trichoderma gamsii</i> ICC080	Soilborne pathogens and grapevine trunk diseases	Competition for space and nutrients mycoparasitism	EU	Radix soil (Isagro), Remedier (Gowan) and others	Gerin et al. (2018), Martínez-Diz et al. (2020)
<i>T. asperellum</i> T25 + <i>Trichoderma atroviride</i> T11	Soilborne pathogens	Competition for space and nutrients, mycoparasitism, antibiosis	EU 2009	Tusal (Certis Europe)	Grondona et al. (2004)
<i>Trichoderma guizhouense</i> CBS 134707 (IBT 41407, G.J.S. 08-135)	Soilborne fungal plant pathogens		USA	Promot WP (JH BiotechInc., Ventura, CA, USA)	Degenkolb et al. (2015)
<i>Trichoderma harzianum</i> + <i>Trichoderma polysporum</i>		Competition for space, mycoparasitism CWDE, antibiosis	Sweden	Binabl [®] (not authorized as BCA in EU)	Khalil and Alsanius (2006)

TABLE 1 (Continued)

Bioactive ingredient(s)	Target (disease or pathogen)	Mechanism(s) and other information	Territories approved/marketed	Product name (supplier)	Reference
<i>T. harzianum</i> T22	Root diseases	Competition in rhizosphere, mycoparasitism, CWDE, antibiosis, induced resistance	USA 1990, EU	Root Shield® (Bioworks), Triatum-P (Koppert)	Blaya et al. (2013)
<i>T. harzianum</i> CBS 134708 (IBT 41408, G.J.S. 08-136)	Soilborne fungal plant pathogens		EU	Vitalin (Vitalin Pflanzengesundheit GmbH, D-Ober-Ramstadt)	Degenkolb et al. (2015)
<i>Trichoderma simmonsii</i> CBS 134706 (IBT 41406, G.J.S. 08-134)	Soilborne fungal plant pathogens		EU	Trichosan® (Vitalin Pflanzengesundheit GmbH, D-Ober-Ramstadt)	Degenkolb et al. (2015)
<i>Serendipita indica</i> (syn. <i>Piriformospora indica</i>)	A wide range of mostly soilborne pathogens	Improves nutrient uptake, but also induces resistance	India	Rootonic: SOM Phytopharma	Shrivastava and Varma (2014), Uma et al. (2017)
Bacteriophage					
Bacteriophage cocktail	Pierce's disease on vine (<i>Xyella fastidiosa</i>)	Parasitism	California	XylPhi-PD, Wilbur-Ellis	Das et al. (2015)
Bacteriophage (presumably a cocktail but not stated)	<i>Xanthomonas campestris</i> pv. <i>vesicatoria</i> , <i>Xanthomonas citri</i> pv. <i>citri</i> , <i>Pseudomonas syringae</i> pv. <i>tomato</i> , <i>Clavibacter michiganensis</i> subsp. <i>michiganensis</i> , soft-rot bacteria of potatoes	Parasitism	USA, Hungary	AgriPhage XCV, AgriPhage-Citrus Canker, AgriPhage PST, AgriPhage CMM, AgriPhage-Fire Blight, Biolyse-BP, Erwiphage	https://www.agriphage.com/product-info/ , https://www.apsbiocontrol.com/products , http://www.erwiphage.com/
Consortium of bacteriophage	Postharvest soft rot of potato	Parasitism	UK	Biolyse-PB, APS biocontrol	https://www.apsbiocontrol.com/products
Consortia	Not specified	Biostimulant	Italy	Coveron, Hello Nature	https://www.hello-nature.com/int/product/coveron-leguminose/
Consortium comprising <i>Glomus intraradices</i> , <i>Funneliformis (Glomus) mosseae</i> , <i>T. atroviride</i> and PGPR					

Note: For USA see also Fravel (2005). CWDE, cell wall-degrading enzyme. PGPR, plant growth-promoting rhizobacteria.

^a<https://cropsscience.bayer.co.uk/our-products/fungicides/serenade-aso/>.

above. Already, in 1972, Jacques and Suoma Ricard founded the firm Binab^R in Sweden producing the *Trichoderma*-based BCA product Binab-TTM and were subsequently among the first to commercialize *Trichoderma*-based BCAs. Now, in 2021, the firm has several products on the market based on *Trichoderma* spp. Several other BCA products from the mid-1980s and 1990s should be mentioned such as MycostopTM, a Finnish product based on a strain of *Streptomyces griseoviridis*, PolygandrumTM—a *Pythium oligandrum*-based product that was also sold in Europe (Vesely, 1989) and in the United States, GliogardTM based on *Gliocladium virens* (syn. *Trichoderma virens*) (Lumsden & Walter, 1996). A more well-known example from the United States came later in the 1990s, when G. E. Harman and two others cofounded TGT Inc., later BioWorks Inc., to commercialize an isolate of *Trichoderma* (T22) originating from the fusion of protoplasts of two different *Trichoderma* isolates (Harman, 2000). Since then, a number of other BCA products have been developed and commercialized worldwide (Table 1); these include bacteria (especially *Pseudomonas* and *Bacillus* strains in addition to the *A. radiobacter* strains) and fungi (especially *Trichoderma* spp. but, for example, *Clonostachys rosea* is also used worldwide).

BCAs identified so far include bacteria, fungi, oomycetes and viruses (Table 1). Successful BCAs have been isolated from soil, especially disease-suppressive soils, as was the case for the parent strains of the BCA T22 mentioned above; they have also been isolated in association with plants, for example, the phyllosphere or rhizosphere, or from within plants, the endosphere. Many of the organisms identified occur naturally in several of these niches. In essence, there is a continuum from soil to rhizosphere (root surface) to endosphere (inside the plant) and phyllosphere (above-ground plant surface) (Jørgensen et al., 2020).

4 | HOW TO FIND A NEW BCA

Two fundamentally different approaches are commonly used in attempts to identify novel BCAs (Figure 1). These are, first, the indirect screening of microbial libraries for antagonistic properties in planta or in silico and, secondly, isolation of organisms from the habitat where the product would be used and then screening them directly for activity in planta (Collinge et al., 2019; Knudsen et al., 1997; Köhl et al., 2011; Teperi et al., 1998). The in vitro approach has been used for high-throughput screening of existing collections of strains for activities against one or more pathogens. We do not know of documented examples of successful products for plant protection from this approach. Direct screening is less suited to high throughput but facilitates the identification of organisms where the mode of action involves plant responses, for example, induced resistance or the ability of an organism to colonize and compete in plant niches (e.g., rhizosphere, phyllosphere, endosphere or in wounded tissue). The advantage of the in vitro approach is that many strains can be tested for the production of antimicrobial metabolites and, for example, mycoparasitic (also termed hyperparasitic) activity. However, both positive and negative results may be misleading as one cannot

be sure that the mechanisms would be active in the plant, nor, conversely, that useful mechanisms are not activated in vitro. The latter has led in many cases to discarding promising BCAs based on in vitro screening (Knudsen et al., 1997; Teperi et al., 1998). There have been many disappointments but a few promising BCAs have been identified (Whipps & Lumsden, 2001). The in planta approach, in its extreme form, involves testing potential BCAs under field conditions. This has been successful for selecting some isolates that are now commercialized (e.g., the product Cedomon; Table 1). However, in practice, it is necessary in most cases to develop tests on plants in growth chambers or greenhouses (Knudsen et al., 1997), or even, in a few examples, on leaves (Latz et al., 2020) or wheat heads (Rojas et al., 2020a) (Figure 2). Although these methods are a compromise, they can simulate conditions that are comparable to the field and can often be carried out throughout the year instead of depending on a brief growth season. Thus, they can give a reasonable level of throughput to select promising candidates for extensive tests in production systems.

Recently, the availability of next-generation sequencing tools has allowed research on biocontrol agents to take a directly functional approach. In *C. rosea* and species of *Trichoderma*, for example, genomics and metabolomics are currently allowing the discovery and investigation of a vast repertoire of specialized metabolic pathways (Karlsson et al., 2015). Study of the roles these metabolites play in the environmental and biotic relations of these organisms may represent a new route to development of BCAs (Mukherjee et al., 2013; Vicente et al., 2020). However, genomic or metabolomics screens are necessarily restricted to looking for signatures derived from study of organisms known to have biocontrol activity. Therefore, such screens should follow on from in planta selection of potential organisms and be a stand-alone approach.

Useful organisms are not only found by targeted searches. For example, a *C. rosea* strain (IK726), originally found in the rhizosphere of a barley root, is effective against many diseases of diverse organs in a wide range of hosts ranging from brassicas to strawberry, oak and cereals (Jensen et al., 2007). Similarly, *Serendipita indica* (syn. *Piriformospora indica*), a plant growth-promoting organism, was found in the root of a desert shrub, but has positive effects for protection against both abiotic stress and attack by certain pathogens in many plant species in very different environments (Cheng et al., 2020; Rabiey et al., 2015; Shrivastava & Varma, 2014). It is commercially available both for biological control and as a biofertilizer (Table 1). In both cases, several mechanisms of action may be operating. Another example is the isolate *Trichoderma gamsii* T6085, isolated from an uncultivated soil in Crimea but effective, when applied on spikes at anthesis, in reducing the incidence of Fusarium head blight on wheat. Like several examples quoted here, it also possesses several quite diverse modes of action, from mycoparasitism to induction of plant defence responses (Matarese et al., 2012; Sarrocco et al., 2013, 2020). Different pathogen lifestyles may necessitate different strategies for identifying and isolating appropriate BCAs. For example, biotrophic parasites of a fungal (or bacterial) pathogen would benefit from the development of methods for isolating and

subsequently cultivating them on bait organisms. This is especially true for viral BCAs that can only live as parasites, for example, bacteriophage (Carstens et al., 2018, 2019; Sabri et al., 2021) on bacteria and mycoviruses and other biotrophic hyperparasites on fungi (Milgroom & Cortesi, 2004; Xie et al., 2011; Yu et al., 2013; Zhang et al., 2020 and Table 1). It can also be a challenge to isolate specialized organisms that may be slow growing or require a host to grow at all; equally, it may be difficult to exploit a slow-growing BCA.

5 | IMPROVED EFFICACY – A KEY TO IMPLEMENTATION?

One of the challenges of biological control is reliable efficacy. Biological control is often considered to be less reliable and efficient than chemical control or host resistance, probably because exposure to the external environment is largely an uncontrollable variable. A counterargument is that some types of biological control (unlike some mechanisms of host resistance) may have an effect against multiple diseases, especially where induced resistance or resistance priming is an underlying mechanism. In addition, it has been shown that, for example, *C. rosea* can be a mycoparasite of diverse fungal plant pathogens such as *Fusarium graminearum* and *Botrytis cinerea* (Jensen et al., 2021). This seems to rely on the response of both general-purpose and specific gene expressions in *C. rosea* depending on which fungal species it parasitizes, indicating that the BCA can work through different modes in biocontrol interactions (Nygren et al., 2018).

Most successful BCAs are effective competitors in the harsh biotic environment of soil and in the plant holobiome (the combination of the plant and its associated microbiome), as they have evolved mechanisms for tolerating toxins from other organisms and are adapted to stressed conditions in those environments, which include growth on roots, stems, leaves and flowers and in wounded tissue. Endophytes, defined as microorganisms colonizing the interior of plants (the endosphere) without causing disease (Jørgensen et al., 2020; Figure 3), are adapted to the ecological niche of the endosphere and are also partly protected from the external environment (and colonize the same niche as pathogens). Therefore, it is suggested that endophytes have the potential to be more consistent as BCAs than purely epiphytic organisms, especially those in the phyllosphere. However, this hypothesis is speculative, based on knowledge that many plant pathogens compete poorly, with an advantage only inside the plant. The hypothesis remains to be demonstrated experimentally for potential endophytic BCAs (Latz et al., 2018). One example is the use of endophytic fungi associated with the invasive weed Japanese knotweed (*Fallopia japonica*). Some endophytes can increase the effectiveness of the rust *Puccinia polygoni-amphibii* var. *tovariae* as a potential control agent of *F. japonica* (Kurose et al., 2012). Another example concerns grass endophytic fungi of the genera *Neotyphodium* and *Epichloë* that can produce alkaloid mycotoxins (e.g., ergovaline) affecting ruminants (especially cattle and sheep). However, some *Neotyphodium* and *Epichloë* isolates can provide a

very high level of protection of the host plant against insect pests (e.g., Argentinian weevil) or fungal pathogens of grasses including *Rhizoctonia* spp., *Bipolaris sorokiniana* and *Curvularia lunata* (Pańka et al., 2013a); *Sclerotinia homoeocarpa* (Clarke et al., 2006); and *Fusarium oxysporum* (Reddy & Faeth, 2010). This appears to be mediated through priming of defences (Pańka et al., 2013b).

Many endophytes only enter the apoplast, but may still have a control function there, either directly inhibiting the pathogens or indirectly by inducing or priming defence responses in the plant (Veloso et al., 2016). However, these organisms might also be adapted to function outside the plant, as is known for *Trichoderma* spp. and *Clonostachys* spp. As good root colonizers, these fungi are also adapted to the harsh environment of the rhizosphere. The isolation of an organism from the rhizosphere or endosphere does not necessarily mean that it can only colonize that environment. Nevertheless, most endophytes will be specialized to some extent to survive inside a plant and would be predicted to compete poorly with microbes outside the plant endosphere. Despite this, there is a continuum in lifestyle, and the same organism may behave as an endophyte, epiphyte or pathogen under different environmental conditions (Jørgensen et al., 2020). Of course, this must be considered in the selection of potential BCAs to prevent accidental selection of plant pathogens.

Consortia, that is, mixtures of microorganisms, are receiving increasing attention as a way of addressing multiple problems. For example, in experiments to control both the fungal pathogen *Fusarium culmorum* and root-feeding insects of wheat, the insect pathogen *Metarhizium brunneum* was combined with the fungal BCA *C. rosea*; effects were observed on both the insect pests and *F. culmorum*, although the efficacy was reduced compared to treatment with either *M. brunneum* or *C. rosea* separately (Keyser et al., 2016). It is tempting to assume that a mixture of BCAs would be more effective than a single agent. However, modelling suggests that, depending on exactly how the organisms compete and act, this may often be untrue (Xu & Jeger, 2013). Different associations can have opposite or antagonistic effects; thus, the ability of *S. indica* to control *R. solani* or *F. oxysporum* infections depended on associated bacteria (del Barrio-Duque et al., 2019). It has also been difficult, except in a few cases, to demonstrate significant additional or synergistic biocontrol efficacy by combining different BCAs in consortia (Xu et al., 2011a, 2011b). One challenge is to ensure that the different agents can operate together under variable environmental conditions and do not have incompatible modes of action. For example, two BCAs acting mostly by bulk nutrient competition would be expected to counter each other's activity. Thus, the idea of forming complex consortia—"synthetic biomes" or "synthetic communities", abbreviated SynComs (Großkopf & Soyer, 2014)—consisting of several different microorganisms with biocontrol effects that could be used as mixtures does not seem to be the most promising route. It can be predicted that there will be selection within a consortium to favour the best adapted to a particular environment and that the dominant consortium members will change following treatment in response to

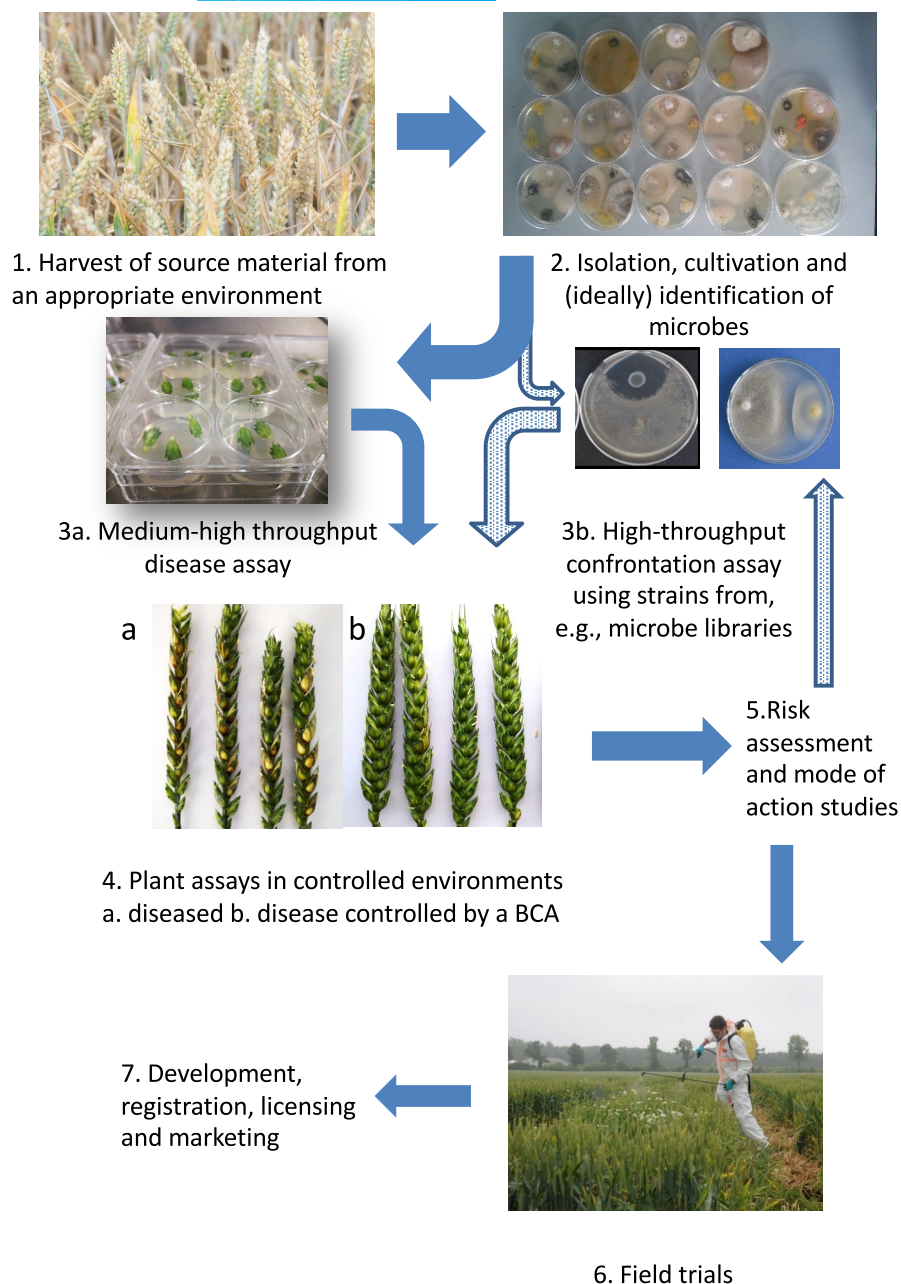


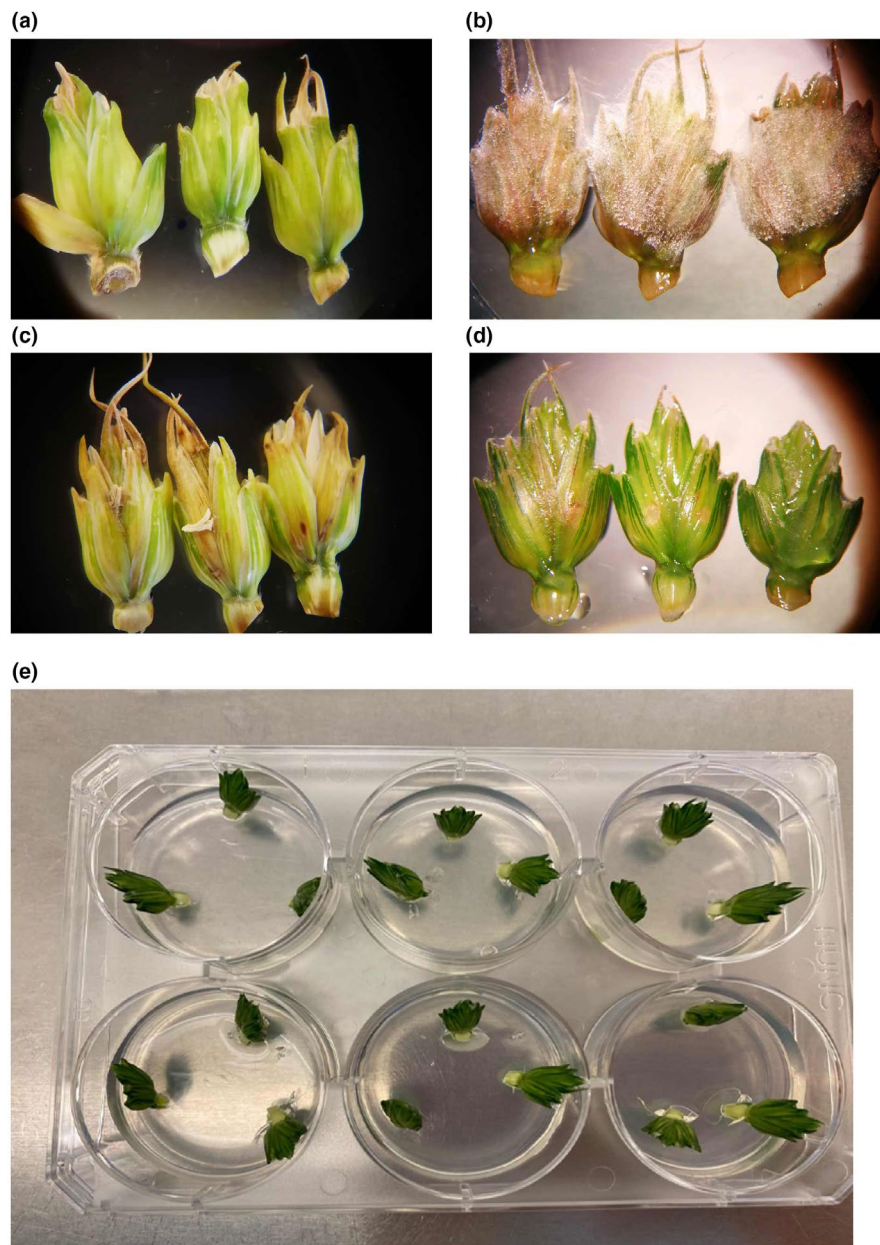
FIGURE 1 Two schemes for selecting potential biological control agents (BCAs)

the local environment. This could be an advantage by making consortia more adaptable to different environments. Nevertheless, in special cases, several products comprising bacteriophage consortia have been released for combating bacterial disease (Table 1).

BCAs are an attractive component in management of postharvest disease, by application at harvest or shortly before. An example is AlfSAFE and similar products for controlling aflatoxin contamination using nontoxigenic *Aspergillus flavus* strains to compete with the toxigenic forms (Amaike & Keller, 2011; Bandyopadhyay et al., 2019). Consumer sensitivity over the use of artificially synthesized chemical application is greater for applications made postharvest than during crop growth; however, the environment is usually less variable or much less variable than in the field, and doses applied can be much more uniform, assisting the use of BCAs that act by resource competition or breakdown of mycotoxins produced by other microbial

species. Nevertheless, biological control using applications of BCAs postharvest is currently not allowed in the EU. Indeed, several products mentioned in Nunes (2012) for European use are no longer approved in the EU, namely, Candifruit™ (*Candida sake*, Sipcam-Inagra, Spain), Pantovital (*Pantoea agglomerans*, Biodurcal, Spain) and Boni-Protect® (*Aureobasidium pullulans*, Bio-protect, Germany). Furthermore, Candifruit™ is considered inefficient (Carmona-Hernandez et al., 2019). In contrast, postharvest BCAs have been used for many years in the United States, for example, to protect soft fruit from postharvest decay before they reach the consumers. *Pseudomonas syringae* ESC-10 is commercialized by Bio-save 10LP in the United States and marketed for several products for postharvest disease control. Examples include use for postharvest control of various fungal pathogens of citrus fruit, pome fruits, cherries and potatoes (product information; Stockwell & Stack, 2007). Postharvest

FIGURE 2 High-throughput assay for *Fusarium* head blight using detached spikelets (Rojas et al., 2020a). (a) Water control, (b) *Fusarium graminearum* (Fg) control, (c) Fg + *Pseudozyma floculosa*, (d) Fg + *Penicillium olsonii*, (e) set-up using large-well plates



BCA treatment of soft fruit for controlling *Penicillium* and *Aspergillus* species and other spoilage pathogens like *B. cinerea* and *Rhizopus* spp. therefore seems to be an important way forward in the EU in view of its successful commercial use in the United States.

Product spoilage can, in some cases, also be avoided by BCA treatments before harvest, depending on the epidemiology of the pathogen–host association. Postharvest problems with mycotoxin production may also be addressed long before harvest by reducing the populations of toxin-producing organisms or the rate at which they produce the toxins, and by increasing the rate and extent that mycotoxins are degraded (Abdallah et al., 2018). For example, mycotoxin production by ear-inoculated *F. graminearum* and *F. culmorum* in wheat was greatly reduced in outdoor (but pot-grown) wheat inoculated with *S. indica* at sowing (Rabiey & Shaw, 2016). This must have been caused by an indirect effect on host resistance because *S. indica* remained restricted to the roots. The doses of BCA culture

used in that study were very large (equivalent to 60 g/m² or 600 kg/ha), but the effect is intriguing. There are interesting examples of beneficial fungi able to degrade mycotoxins, for instance, the ability of *C. rosea* to detoxify the mycotoxin zearalenone (ZEA) through the enzyme zearalenone lactonohydrolase has been demonstrated (Kosawang et al., 2014) and there are promising results from the field where *C. rosea* has reduced the deoxynivalenol content in harvested wheat grain (authors' unpublished data). Similarly, the ability of some *Trichoderma* isolates to degrade mycotoxins has recently been studied. In the case of *Trichoderma aggressivum*, its zearalenone lactonohydrolase was expressed in *Escherichia coli* BL21 (DE3) and successfully purified (Chen et al., 2021).

Postharvest pathogens on soft fruit such as mycotoxin-producing species of *Aspergillus* and *Penicillium* are not likely to be controlled efficiently preharvest, even though it is often suggested that application of beneficial organisms preharvest can reduce mycotoxin

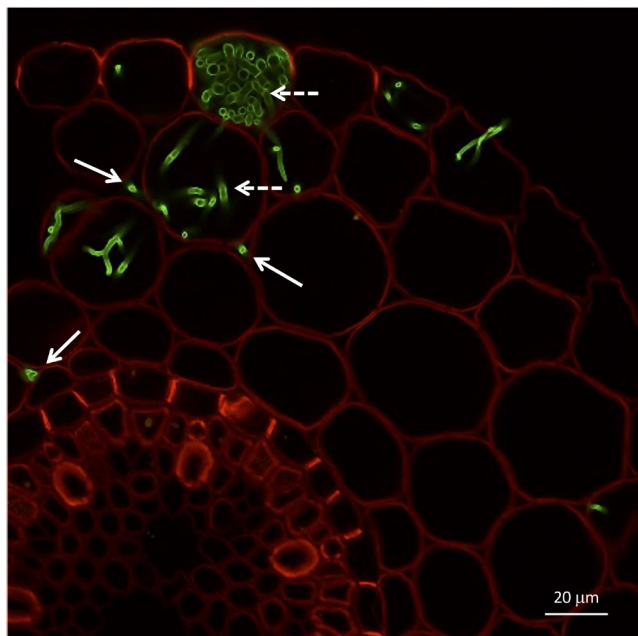


FIGURE 3 Endophytic colonization of wheat root by *Trichoderma gamsii* T6085 7 days postinoculation: arrows indicate intracellular (dashed line) and intercellular (continuous line) colonization by *T. gamsii* T6085 hyphae. Fungal cells were detected with WGA-Alexa Fluor 488 (green channel); the plant cell wall was detected with FM4-64 dye (red channel) by confocal microscopy. (Photography: Sabrina Sarrocco & Marie Dufresne)

accumulation postharvest (Sarrocco & Vannacci, 2018). There are exceptions. This is the case for beneficial yeasts such as *A. pullulans*, whose preharvest application on grape resulted in a reduction of ochratoxin A contamination by around 95% (Dimakopoulou et al., 2008). Another interesting example is *Kluyveromyces thermotolerans*, able to control the growth of *Aspergillus carbonarius* and *Aspergillus niger* in the field by up to 100% and to reduce mycotoxin accumulation by up to almost 80% (Ponsone et al., 2011).

6 | MECHANISMS – MODES OF ACTION

There are four main modes of action underlying biological control of plant diseases (e.g., Jensen et al., 2017): (a) exploitation competition for resources (oxygen, carbon, nitrogen and other essential resources); (b) interference competition for space via antibiosis where the BCA inhibits the pathogen via effects of toxic secondary (specialized) metabolites; (c) hyperparasitism, where the antagonist acts as a predator and exploits the pathogen as a prey; (d) induced resistance—the indirect interaction of a BCA via induction of plant defence mechanisms against invading pathogens. A fifth mechanism that can contribute to disease control is plant growth stimulation by better nutrient absorption and/or by affecting plant hormone pathways, as demonstrated by, for example, various rhizosphere bacteria and fungi. A strongly growing plant may be better able to withstand a pathogen and a rapid establishment of seedlings in the field can

lead to avoidance of damping-off diseases. However, some researchers would not consider this as biological control, as discussed earlier in this review (Stenberg et al., 2021).

A single BCA may exhibit a combination of these modes of action. The individual modes of action have different but not exclusive population dynamic consequences. It can be quite difficult to prove that a particular mechanism is operating in planta, even though it can operate in vitro (Latz et al., 2018). More than one of these mechanisms can contribute to a concerted action in a particular case and the importance of a specific mechanism used can vary from case to case, even using the same organism. For example, species of *Trichoderma* and *Clonostachys* may act as hyperparasites, metabolite producers, competitors and/or modulators of plant defence responses (Benítez et al., 2004; Harman, 2006; Jensen et al., 2021; Mukherjee et al., 2013). Exploitation competition can be independent of the pathogen population size, simply reflecting efficient local resource capture. Competition through more efficient resource use does not rely on direct interaction as the BCA takes over resources and space meaning the pathogen cannot thrive. Being the first to colonize new resources is another important type of exploitation competition that can deprive a pathogen of the resources it requires, especially in the critical early stages of colonization. In addition, the ability of beneficial organisms to colonize a substrate that is not preferred by the targeted pathogens could improve competitiveness of the biocontrol agent against the pathogen in the targeted pathogen community (Lasinio et al., 2021). Alternatively, interference competition through antibiosis, depending on how close the organisms need to be to interact, may allow the BCA to monopolize the habitat (Sarrocco et al., 2019). Hyperparasitism requires that the BCA occurs and is metabolically active spatially close to the target pathogen (normally in the niche where the pathogen would infect, or which is occupied by fruiting bodies or resting structures of pathogens that are parasitized by a BCA).

The question was raised in the webinar whether pathogens could evolve to be resistant to BCAs, as frequently occurs with repeated use of pesticides with specific modes of action. Over more than four decades of using biological control, resistance in the target bacterial and fungal pathogens has yet to be demonstrated. The direct use of metabolites and extracts, which is not included in the definition of biocontrol discussed earlier, leads to high exposure of the pathogen; this may carry more risk of development of pathogen resistance, as seen with the regular use of chemical pesticides. In the case of bacteriophages, it is known that bacteria can adapt rapidly to bacteriophages and are expected to overcome single strains. Products are therefore based on cocktails of bacteriophages to reduce this problem (see below).

Although resistance has not been considered a serious problem for most other practical uses of biocontrol, we will next discuss the issue and its relation to mode of action. It is not easy to see how a pathogen could evolve resistance to exploitation competition in nature. However, as for chemical pesticides, resistance towards BCA metabolites in pathogen populations is a theoretical possibility; if a substantial proportion of a pathogen population is regularly

exposed to this metabolite, leading to high selection pressure, resistant phenotypes could, in principle, arise. Some BCAs mainly rely on antibiosis via the production of one or a few specific toxic metabolites; resistant pathogen phenotypes could result, perhaps with a consequent risk of field resistance. An example of one stage in this process has been observed in take-all decline of wheat, induced by monoculture, in suppressive soils. Isolates of the pathogen involved (*Gaeumannomyces tritici*) showed variation in sensitivity to two metabolites produced by strains of *Pseudomonas fluorescens* that were claimed to be important for disease suppressiveness (Mazzola et al., 1995). Such variation in different traits is to be expected but, based on the studies by Mazzola et al. (1995), there is no clear evidence that the population as a whole has become less sensitive to the two metabolites tested (phenazine-1-carboxylic acid or 2,4-diacetylphloroglucinol) despite heavy exposure to them. Furthermore, no evidence of resistance to 2,4-diacetylphloroglucinol was found in *G. tritici* populations from Washington State, USA (Kwak et al., 2009).

In general, pathogenic organisms can be expected to vary in traits allowing them to thrive in variable but competitive environments (Dubey et al., 2014; Karlsson et al., 2015). Because resistance to a metabolite can be conferred by changes in the target site, detoxification, excretion (efflux) or general metabolic adjustments, intensive use of a BCA acting via antibiosis and based on one or a few specific toxic compounds could lead to the evolution of resistant pathogens. The selection pressure is increased if pathogen populations experience heavy (long term and/or highly effective single dose) exposure to the metabolite. For this reason, vulnerability to resistance should be considered on a case-by-case basis when creating strategies for biocontrol use. There is a strong argument for the development of many different BCAs for a given problem, to avoid exposure of large proportions of the pathogen population to the same selection pressure.

A special case where a strategy for avoiding resistance in pathogen populations has been addressed is the biocontrol of crown gall, caused by the bacterium *A. tumefaciens*, using the BCA *A. radiobacter* (syn. *R. rhizogenes*) strain K84 that produces the toxin agrocin responsible for the antibiosis (reviewed by Penyalver et al., 2000). Here, the BCA harbours a plasmid that encodes resistance to its own agrocin toxin and at the same time encodes mobility of this plasmid with resistance to other *Agrobacterium* strains. In this case, the concern was that the plasmid might be transferred to the plant-pathogenic *Agrobacterium* bacterial strains making them resistant to agrocin. As this was demonstrated to happen both in field and in laboratory experiments and information accumulated that it also might be happening under commercial use, a gene modification of the BCA was created in which the plasmid mobility trait was deleted—strain K1026. This strain K1026 has been used commercially in Australia and in the United States, although biocontrol with commercial use of the wild-type strain K84 still provides effective biocontrol in many crops worldwide after almost 50 years of commercial use (Kerr & Bullard, 2020). Strict legislation for regulating BCAs has until now prevented the use of *R. rhizogenes* for biocontrol in the EU, but both

the mutant K1026 and the wild-type K84 are approved in many other countries (Kerr & Bullard, 2020).

A specific (biotrophic) hyperparasite requires a pre-existing host population to parasitize as well as a living host for activity and growth; therefore, they will be effective in the short term only if applied inundatively. An exception may occur if a biotrophic hyperparasite could function effectively and survive in the longer term in an environmental reservoir. Unfortunately, such biotrophic hyperparasites will not usually compete well in the absence of a host (Bennett et al., 2003). Nonetheless, there are some examples of commercialized biotrophic hyperparasites used for biocontrol such as *Ampelomyces quisqualis* (Figure 4), used against powdery mildew (Karlsson et al., 2018) and *Coniothyrium minitans*, a parasite of several sclerotia-forming plant pathogens (Whipps et al., 2008). A special example of a potential BCA is the hyperbiotrophic fungus *Pseudozyma flocculosa* (a yeast) that parasitizes powdery mildew and, in this way, obtains access to resources from the leaf infected by the mildew fungus. *P. flocculosa* is dependent on a living host–pathogen combination and so needs to find a new host mildew if a mildew colony dies (Laur et al., 2017). Interestingly, *P. flocculosa* also produces an antifungal glycolipid, flocculosin, suspected to have a role in the interaction. However, a CRISPR-Cas9 mutant impaired in the biosynthesis of flocculosin was apparently unaffected in its ability to antagonize powdery mildew (Santhanam et al., 2021). This is an effective lifecycle as powdery mildews are polycyclic, the organism attacks multiple species of powdery mildew, and new infections are found throughout the growing season in many crops, continually offering new living hosts for the BCA.

Whether the use of specialized hyperparasitic BCAs would be risky in an inundative strategy should be considered case by case. It is possible to set up an effective strategy for their use, provided knowledge of the target pathogens and their disease cycles, the environmental conditions, the biology and ecology of the BCA allow the prediction of the right timing and placement of the BCA in the niches where it is to act. *A. quisqualis*, for example, is effective against powdery mildew on cucumber (Sundheim, 1982) but less effective in controlling powdery mildew on grapevine caused by *Uncinula necator*, as it mainly parasitizes the fruiting bodies (chasmothecia) late in the season (Falk et al., 1995a, 1995b). Parasitism of the conidial stage throughout the growing season is highly dependent on humidity, which is not a requirement for conidial production by the pathogen. Therefore, the BCA is less efficient in periods with low rainfall/humidity. However, as Falk et al. (1995a, 1995b) point out, parasitism of chasmothecia might have an important role in integrated disease control by reducing primary inoculum for the following year.

Although not a crop example, the rust hyperparasite *Sphaerellopsis filum* appears to have specific genotypes that are adapted to attack only some genotypes of individual species of grass-infecting rusts (Kajamohan et al., 2015); this phenomenon might also be relevant to other biotrophic hyperparasites. Viruses can also be considered as obligate hyperparasites with more or less specific host ranges (see below).

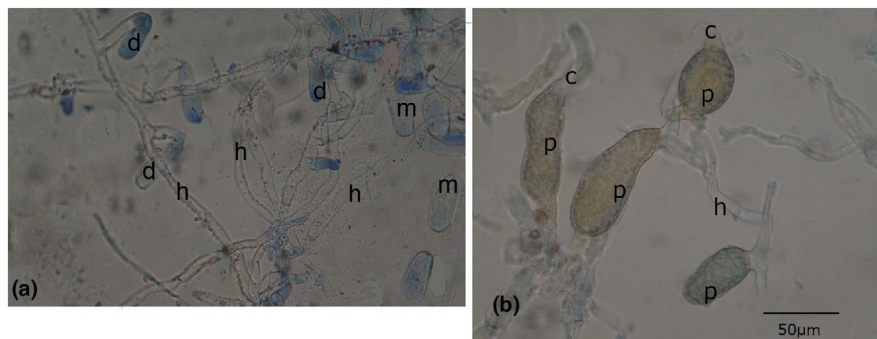


FIGURE 4 (a) Healthy powdery mildew colony on courgette (zucchini, *Cucurbita pepo*) leaves. h: hypha; d: developing conidium on conidiophore; m: mature conidium (b) *Ampelomyces* sp. growing on the mycelium of powdery mildew and suppressing conidial production. p: pycnidium; h: mildew hypha; c: tip of mildew conidiophore. Note the absence of mildew spores: all mildew conidiophores are surrounded by *Ampelomyces* pycnidia. (Photography: Michael Shaw from surface strips on transparent sticky tape; pictures edited to remove air bubbles; scale bar applies for both images, approximate, for guidance only)

However, most BCAs that work via hyperparasitism are necrotrophic parasitic fungi that compete well and survive without a living host pathogen. Examples are species of *Trichoderma* and *Clonostachys* that can operate as mycoparasites as part of their life-style but also grow and multiply via other ways of life, as addressed in more detail in Karlsson et al. (2018). Necrotrophic hyperparasites are considered more aggressive as BCAs than the more specialized hyperparasites, and are more competitive, for example, in the rhizosphere and in root colonization.

Induced resistance is a well-studied phenomenon in the laboratory where there are good examples of this as a mode of action. However, induced resistance will be ineffective against existing high population densities of a pathogen. Interaction with target pathogens via induced resistance does not require close proximity of the target and the BCA. For example, root application of *S. indica* can stimulate both plant growth and induced resistance in the shoot (Ntana et al., 2022; Rabiey et al., 2015). Volatile specialized metabolites can act as signals between plant parts and, at least in principle, between neighbouring plants (Farag et al., 2013). Moreover, application of BCAs can induce resistance in the progeny of treated plants (Medeiros et al., 2017), a phenomenon termed “transgenerational systemic acquired resistance” (Luna et al., 2012). Several phytohormones have been shown to be involved in the resistance induced by *S. indica* (Hilbert et al., 2012; Jacobs et al., 2011; Khatabi et al., 2012). Hormones have complex and sometimes antagonistic effects, which can influence the response to both abiotic and biotic stress, modifying cellular physiology to respond and adapt to the stress. For pathogens, the activated defence responses provide induced resistance (PAMP-induced immunity; Ray et al., 2018).

Understanding the evolutionary response to the use of host resistance inducers raises the question of why plants do not trigger these defences constitutively. The obvious answer is that induced resistance requires energy or involves intrinsic damage such as cell death, and so there is a fitness cost. This means that the induced defences are regulated (e.g., by transcriptional modulators such as NPR1) and not deployed unless needed. In this case, therefore, the use of a BCA to induce resistance in the absence of a substantial

subsequent pathogen attack could lead to loss of yield. This would be a serious set-back in developing BCAs as part of an integrated disease management toolbox. Although negative effects of application in the absence of pathogens are hard to demonstrate, experiments involving transgenic plants where constitutive expression of *R* genes (Oldroyd & Staskawicz, 1998) and regulators such as *Npr1* (Backer et al., 2019; Silva et al., 2018) were used can result in enhanced induced resistance with demonstrable fitness costs (Collinge et al., 2010). One of the great challenges for the genetically modified organism (GMO) approach in recent decades has been the identification of appropriate promoters for driving the expression of such genes. The use of tissue-specific promoters can mitigate the negative effects of inappropriate expression (Tripathi et al., 2016).

The effect of *S. indica* (and some other agents) has been suggested to be first and foremost growth promotion (Gill et al., 2016) allied to effects such as drought tolerance. In that case, the question remains of what prevents evolution of constitutive expression of the growth-promoting traits. Apparently, defences can be, if not activated, primed, with effects on growth and yield that are too slight to measure (Conrath et al., 2006). There are two hypotheses that could explain why the defences remain facultative: (a) the costs are expressed in specific circumstances, not usually encountered in experimental or field-crop settings; (b) less probably, it may be that in natural settings, with a diverse and microbe-rich soil, priming always happens, so there is no selective advantage or disadvantage in facultative control—it is just a normal stage in development. If (a) is correct, there is the important practical conclusion that we should be looking very hard for side-effects of these priming organisms before they are too widely deployed on crops.

7 | ENVIRONMENTAL MANIPULATION AND SUPPRESSIVE SOILS

Environmental manipulation is often used as an approach to achieve biological control against insect pests, such as the promotion of bio-diverse crop margins to encourage predators, providing biological

pest management under the title of Conservation Biological Control. This is used rather less against pathogens. Reduction in attack by pathogens can be achieved, in principle, by manipulating the habitat to encourage one or more BCA in the soil, or perhaps by using adjacent vegetation to encourage the right individual microorganisms or microbiomes. The use of elemental sulphur to lower the local pH and discourage *Streptomyces scabies*, causing scab on potato, is perhaps an example (Vlitos & Hooker, 1951). Another example is watering potato plants during tuber formation to stimulate colonization of the new lenticels by antagonistic bacteria (Cook & Papendick, 1972; Ryan & Kinkel, 1997). Similarly, later in the season, damage from eyespot of wheat caused by *Oculimacula* spp. was—counterintuitively—reduced by ceasing straw burning (Jalaluddin & Jenkyn, 1996). Compost and especially “compost tea” may provide a source of BCAs or alter the nutritional environment to favour BCAs, which are responsible for the activity of the compost tea (St Martin, 2015). Biochar is hypothesized to provide increased surface area suitable for microbial growth and may interact desirably with compost teas (Edenborn et al., 2018). These approaches are a subject ripe for study, though reliability has been a major problem; metagenomic and community metabolism methods may improve this (Edenborn et al., 2018). Part of the effect of good cultural practices—though perhaps unconsciously—is likely to be the encouragement of microbial communities that either prime or induce plant defences, or act as direct BCAs.

Microbiota can increase natural soil suppressiveness against soilborne pathogens particularly when intensive cropping systems (with high inputs of synthetic chemicals, low soil organic matter accumulation, little humification and frequent soil tillage) are the primary reasons for soil depletion (Cook, 1992). Soil microbiota associated with biocontrol can be a key factor in the beneficial influence of agronomic practices on plant health (French et al., 2021). Next-generation sequencing often offers a deeper characterization of the soil microbial community during microbiome manipulation. This may allow a more mechanistic understanding of what is happening and the effect on crops in terms of soil suppressiveness, so helping to limit inconsistencies, drawbacks and failures related to soil microbiota disturbance (De Corato, 2020). More generally, the “omics” sciences—through a combination of metagenomics, meta-transcriptomics, meta-proteomics and metabolomics approaches—should help in understanding the microbial activities as a whole and the potential of the plant-associated microbiota to suppress plant disease (De Corato, 2020; Schlatter et al., 2017).

8 | HOST GENOTYPE AND PLANT BREEDING

Another exciting research area related to biocontrol is the interaction between plant host genotype and microbiome. Just as disease resistance is inherited, it is predictable that the microbiome of a plant, which is relevant to biological control activity, is affected by genotype. We can predict that deepening knowledge of how

agronomically important traits relate to plant function will increasingly contribute to our ability to predict the effects of genotype variation on responses to BCAs. For example, the effect of *S. indica* on wheat response to drought stress is strong but variable, with quantitative trait loci having large apparent effects (Amer, 2020). It seems very likely that genotype would also affect the control of *Fusarium* spp. on the crown and ears shown in previous work (Rabiey & Shaw, 2016; Rabiey et al., 2015). While this would complicate management, the consideration of diverse varietal susceptibility to multiple diseases is routinely part of farm decision making.

Another factor that plant breeders should consider is the genotype of the host and native microbiome. Some *Trichoderma* isolates, by endophytically colonizing host roots and shoots, establish a molecular dialogue resulting in desirable effects on plants (Macías-Rodríguez et al., 2018; Ramírez-Valdespino et al., 2019). This phenomenon was first described in 1952, when Mostafa and Gayed (1952) reported that *Trichoderma* improves fresh and dry weight in cotton plants. More than 20 years later, exudates from lettuce were reported to have a beneficial effect on germination of *Trichoderma viride* conidia, indicating that fungus and plants obtain mutual benefits (Catskä et al., 1975). What was not known, and indeed drove researchers to more basic studies, was that the beneficial effects of *Trichoderma* application depend on the plant genotype. This concept has been proven in the interaction between *Trichoderma harzianum* T22 and maize (Harman, 2006) and between *Trichoderma* and tomato (Tucci et al., 2011). However, the influence of the microbiome must be studied on a case-by-case basis; a recent study on wheat looked at the endomycobiome (i.e., fungal endophytic microbiome) of wheat but could find no relation to resistance to *Zymoseptoria tritici* (Latz et al., 2021). In contrast, Mahoney et al. (2017) observed that wheat cultivars may consistently alter the rhizospheric bacterial community, thus providing beneficial services to the host. Plant genotype, including hosts already affected by a disease, seems to play a crucial role in the recruitment of rhizosphere bacterial microbiota, at least in a controlled environment, an approach suggesting the need for further investigation into soilborne plant disease suppression (Dilla-Ermita et al., 2021; Yin et al., 2021).

9 | RISK ASSESSMENT

Just because something is “natural” does not mean that it is “safe”. For approval, BCAs have to be assessed for potential harmful activities to farmers and consumers, and for negative effects on the environment and other crops. Several categories of risk need to be considered before a BCA (or any other novel product) can be considered reasonably safe for possible commercialization or recommendation (Ehlers, 2011; Sundh & Eilenberg, 2021). Screening and isolation of new organisms concentrate on looking for promising organisms before worrying what they are. However, by the early stage of serious screening programmes it is already necessary to identify the organisms that are being selected as potential BCAs. This is to avoid selection of plant pathogens, human and/or

farm animal pathogens, or mycotoxin-producing strains. Aspects to consider when starting a screening programme are discussed by Köhl et al. (2011).

A few examples of potential BCAs, when finally identified, have turned out to be possible human pathogens. For example, a bacterial strain that had good activity against *Didymella bryoniae* was isolated from watermelon roots but was found to be the human pathogen *Pseudomonas aeruginosa* (Nga et al., 2010). The *Burkholderia cepacia* complex, defined by Eberl and Vandamme (2016) as containing “good and bad guys”, includes several BCAs of plant diseases and has been actively exploited in bioremediation. However, because the *B. cepacia* complex also contains strains described as plant pathogens or opportunistic pathogens of humans affected by cystic fibrosis, the U.S. Environmental Protection Agency reassessed the risk of several isolates already registered for biological control (Parke & Gurian-Sherman, 2001). Another risk is, as mentioned above, the production of harmful metabolites or even mycotoxins by a successful BCA. The ascomycete *Chaetomium globosum* can control the serious apple pathogen *Venturia inaequalis* of the phyllosphere but its production of toxins led to it being abandoned as a commercially viable BCA (Boudreau & Andrews, 1987). The greatly reduced costs and improved efficiencies in genomic sequencing over the last decade provide excellent opportunities to avoid this type of unpleasant surprise. It is to be expected that plant pathogens will be isolated and enter into the first stage of screening for potential BCAs, because the sources of promising microorganisms will often be plant biomes including endophytes (Latz et al., 2021; Manzotti et al., 2020; Rojas et al., 2020b). However, a universal exclusion of possible BCA candidates based on their species-level taxonomy risks missing useful organisms. For example, fungi within the species *F. oxysporum* can be grouped into either nonpathogenic or pathogenic individuals. Those belonging to the pathogenic group can again be subdivided into *formae speciales*, depending on the specific host plant they can infect and cause wilt disease in. Indeed, nonpathogenic *F. oxysporum* strains are promising BCAs derived from disease-suppressive soils (Alabouvette, 1986). These strains are, for example, good at controlling wilt in tomato caused by *F. oxysporum* (Alabouvette et al., 2009) or *Verticillium albo-atrum* in pepper (Constantin et al., 2019; Veloso et al., 2016). The basis of host range among pathogenic strains in *F. oxysporum* has been shown to reside on supernumerary chromosomes (Ma et al., 2010). Similarly, the acquisition of *ToxA* from *Parastagonospora nodorum* by *Pyrenophora tritici-repentis* (Friesen et al., 2006) has led to serious new disease problems. Would it be possible to ensure that a successful BCA could not gain a chromosome or chromosome segment and become a pathogen of its present host or of other crops? This scenario seems, fortunately, to be rather unlikely as, for example, nonpathogenic *F. oxysporum* coexist naturally with the pathogenic strains and with other species of *Fusarium* in many soils, apparently without leading to new pathogenic strains. Furthermore, such transfer of pathogenicity has not been observed in augmentative biocontrol experiments with nonpathogenic *F. oxysporum*, although clearly the process cannot be totally ruled out.

For some BCAs, perhaps particularly for species operating by induced resistance, there is also a risk that weedy hosts might be made more competitive by interaction with the BCA, particularly if it has a wide host range. For example, *S. indica* improves the growth of many wheat cultivars, as mentioned above—but also has, as do other *Serendipita* spp., beneficial effects on some competing weeds (Edenborn et al., 2018; Rabiey et al., 2017; Ray et al., 2018). More research is needed to clarify whether this could really be a problem in crop production.

Whereas a BCA needs to be sufficiently aggressive to be active against its target without use of uneconomic volumes or numbers of applications, we should also be able to recover from unexpected ecological or medical effects. This leads to the argument that an agent should not persist too long in the environment. Commercially, the advantage of this is that the product has to be sold every year, allowing recovery of the research investment over a long period. Perhaps an average of one growing season should be enough? Is it ethical to develop BCAs that can persist and become permanent components of the local microbiome or would this be a godsend for agriculture—if they do not spread to natural habitats and change ecosystems? For perennial plants, would it be sufficient to ensure that they do not spread from the inoculated host? This requirement, of course, is in conflict with the desire to be able to encourage BCAs in the environment by habitat manipulation.

There is a political movement to speed up the process of approval for BCAs, on the probably spurious grounds that they are intrinsically safer than artificially synthesized molecules. For instance, in the EU, where the process is considered to be as painstaking as for new pesticides or GMOs (Sundh & Eilenberg, 2021), the argument has been made that strains closely related to existing approved products should not need the same level of documentation before being licensed for release. Of course, there would still be risk, and some kind of “yellow card” system, like that used to report possible side-effects of medical interventions, would be desirable.

However, in some countries, most prominently in Brazil, people from farms are starting to use home-grown biomass of beneficial isolates (such as *Trichoderma* spp.) in order to have the quantity required to treat their fields. Without being supported by adequate facilities and without a basic knowledge of the organisms the growers are managing, the risk of contamination of the target strains is likely. The products applied to crop fields could therefore be completely different from the original strains, with the consequences that (a) any kind of beneficial effect is reduced or eliminated; and (b) the supposed BCA could be dangerous for the producers and the consumers of the final product. Lastly, but of no less importance, almost 90% of a BCA product is usually represented by coformulants that guarantee the survival and quality of the active BCA, and therefore assure good disease control (Lana et al., 2019). The correct mix of coformulants cannot be expected to be reproducible in home-made BCA products. Strict regulation is needed in these countries in order to reduce the risks connected with this trend and to ensure that the products sold actually work and are not just harmless—or even harmful—mixtures. Quality control is vital to achieve effective

biological control, but home-made products, including compost teas, cannot be controlled for consistency and safety.

10 | LEGISLATION AND REGISTRATION

Factors that are considered in the approval processes around the world include production of toxic metabolites, pathogenicity to humans or crops, allergenicity and ability to survive and spread. Some countries have very little regulation whereas other regions (EU, USA) impose strong constraints on the documentation for safety and—in the EU but not the United States—efficacy, before permitting commercialization. The challenges regarding registration of BCAs were the focus of a white paper from the EMPHASIS project (EMPHASIS, 2016), which called for more harmonization, as did a workshop in the same year convened by the International Organization for Biological Control and summarized in Ward (2016). Another important recommendation was that benefits as well as risks need to be taken into account when considering biocontrol agent release permissions. The current system of Pest Risk Analysis only focuses on the latter.

As mentioned above, the first GMO product was the strain K1026, modified from *A. radiobacter* K84, originally marketed as NOGALL® in Australia (Kerr & Bullard, 2020). It is interesting to consider other categories of BCA and how their use and regulation has evolved alongside agents for disease management. BCAs intended for reduction of weed populations can be considered to be “classical”—agents that are expected to offer long-term reduction in target populations, without repeated widespread release—or as mycoherbicides, requiring regular and widespread release. The regulatory requirements differ.

In the case of classical weed biocontrol agents, the focus in the early part of the 20th Century was on safety to crops and little else. Subsequently, protecting native species became politically important and a thorough risk assessment is now demanded prior to the release of any BCA active against weeds. This includes centrifugal (testing close relatives first) host specificity testing based on plant phylogeny and typically involves 50–80 species of nontarget test plants being exposed to the potential agent, be they fungal or arthropod. However, with the advent of molecular tools to determine phylogeny better, much shorter test plant lists are proposed (Briese, 2003). This level of investigation normally satisfies the licensing authority of recipient countries, most of which have legislation banning the introduction of non-native organisms. In the United Kingdom, permission to release arthropods is gained through the Wildlife and Countryside Act 1981, often with input from the Advisory Committee on Releases to the Environment, a public consultation and ministerial approval. The fact that weed biocontrol agents are classified as “likely to be injurious to plants in the United Kingdom” puts them under Plant Health Regulation (Shaw et al., 2016; Box 1).

If considering developing a mycoherbicide, the registration process, at least in Europe, is the analogue of registering BCAs for plant disease control and the chemical pesticide registration process; this is often cited as a reason for the slow development and poor pipeline

of alternative products for pest and weed management (Bale, 2011; Zaki et al., 2020).

In the case of classical arthropod biocontrol, the restrictions are technically the same as agents targeting weeds. As the predators and parasitoids are not plant pests, there are no plant health quarantine restrictions placed on the research, but responsible researchers would take precautions to prevent escape prior to licensing. The level of host range testing applied to insects intended for insect biocontrol is rather less than for weeds. In Europe, many of the 176 species of arthropod BCAs have been released outside the glasshouse without much host range testing or risk assessment at all. The on-going and catastrophic invasion of the intentionally released predatory harlequin ladybird, *Harmonia axyridis*, shows how significant nontarget impacts can be when things go wrong (Kenis et al., 2017; Roy & Wajnberg, 2008). Nonetheless, extensive analyses have demonstrated that nontarget effects impacting native species at the population level are rare when compared with the number of introductions that have occurred (Hajek et al., 2016). As with BCAs targeting pathogens, there are conflicting advantages to modes of action: parasitoids are (sometimes very highly) specialized, which makes them less attractive for commercialization and more vulnerable to counter-evolution; in contrast, predators have a wider host range but with correspondingly greater danger of unexpected ecological damage (Louda et al., 2003; reviewed by Taylor & Snyder, 2021).

11 | VIRUSES AS MANAGEMENT TOOLS AGAINST BACTERIA AND FUNGI

A form of hyperparasitism that is receiving increased attention as a new approach to biological control is the use of viruses to infect and weaken fungal or bacterial plant pathogens. The potential to use mycoviruses for controlling chestnut blight caused by the ascomycete *Cryphonectria parasitica* has been studied for decades and is effective (so far) in some regions but has not proved sufficiently effective in others (Milgroom & Cortesi, 2004). A more recent and very promising example concerns a mycovirus with a 2 kb genome that converts the necrotrophic fungus *Sclerotinia sclerotiorum* into a beneficial BCA that induces resistance and the mycovirus can also infect and inactivate other pathogenic strains it meets (Zhang et al., 2020).

Bacteria are difficult to control other than by cultural practice and disease resistance, if available. Recent studies suggest the potential for bacteriophages to control bacterial diseases (Ahern et al., 2014; Carstens et al., 2019), and indeed, the first product, which is against Pierce's disease and is based on a cocktail of four bacteriophages, has now reached the market (Table 1) (Das et al., 2015). The use of bacteriophages to control human disease has been explored since their discovery (Abedon et al., 2011; Furfaro et al., 2018; Sybesma et al., 2018). A challenge with bacteriophages is the need to prepare mixtures of bacteriophages specific to each of the component genotypes in the mixture of host bacterial types causing a problem. This also means that resistance is likely to be a major

and rapidly arising problem because of the naturally occurring host-bacteriophage coevolutionary race that indeed underlies the initial need to use mixtures. Thus, there are two points here: (a) specific matching for effectiveness, and (b) the complications of the evolutionary process driven by host-bacteriophage matching. The need to use tailored mixtures was an important reason why bacteriophage therapy for humans has developed slowly in Western medicine—the Soviet bloc used it but needed to maintain large banks of bacteriophages against every subtype of bacterium they were trying to control. Though this sounds complicated, the positive side of this is good control of use, because the bacteriophage cocktails used have to be compiled according to need and resistance management can be built in. The negative side is the potential for erratic severe outbreaks. The same considerations apply to mycoviruses. Alternatively, in many of these cases, the narrow host range can be considered to be a biosafety advantage, though there can be advantages in broad host range (Ross et al., 2016).

12 | COMMERCIALIZATION

With high development costs and limited targets there have been relatively few market successes and the availability of specific products is often restricted to one or a few countries or limited regions (Table 1) (Cordeau et al., 2016). Few BCAs are as effective as established pesticides. Thus, the market opportunities occur where a gap in activity opens due to consumer choice, safety issues or the evolution of pesticide resistance. However, it seems that biological control can play important roles in integrated pest management strategies for reducing input of chemical pesticides and in organic plant production.

In general, commercialization of a BCA is very challenging and many potential products are never brought to the market. The challenges to successful commercialization of a BCA are many and range from developing the biological production process to raising the capital required for manufacture, distribution and successful marketing (Table 2). As we have already discussed, good control by BCAs has been achieved many times in controlled environments and artificially simplified ecosystems but it has often proved difficult to translate these achievements to commercial or other agricultural settings, whether field, greenhouse, forest or plantation. This is not surprising, because we know that the severity of disease caused by pathogens is subject to environmental influence by factors such as humidity and temperature (the “disease triangle”). Biological control represents adding a third living organism, with its own environmental envelope, to the system. BCAs can be applied in many ways, such as spraying, application to planting material (e.g., seed coating), soil surface mixing, postharvest spray or aerosol application. Determining optimal formulation of a living organism, choice of mode of deployment, and design of field trials are also challenges prior to commercialization. Once these issues, including registration, have been solved, there are the problems of being able scale up to a profitable production level; the product must be reliable and

add sufficient yield and/or quality to the crop to give net profit to a grower and, ideally, be sufficiently nonspecific to allow development costs to be spread over multiple targets. Shelf life is perhaps not a major issue in industrialized agriculture but is clearly a concern for rural communities in developing countries. How BCA stability compares to that of chemical pesticides is also an important question. Despite these challenges, there are a number of European examples of successful commercialization of biological control products (Table 1) and there are now many companies within the agroindustry that are aiming to market new BCA products. A prospect that we do not address here is the possibility of combining fungicides and BCA in integrated management.

The global BCA market, continuously increasing, reached almost \$4.0 billion in 2020 with a projection towards \$10.6 billion by 2027 (Anon, 2020). Several governments are supporting the use of more environmentally friendly agri-inputs, especially when we gradually recover from the COVID-19 pandemic. North America, under stringent rules and regulations regarding the use of chemical crop protection products, is currently the largest market for BCAs and this is expected to continue throughout the forecast period. South America is particularly promising as a market, with Brazil and Argentina showing an increase of area under organic farming (Paull & Hennig, 2019) and a consequent rise in demand for BCA products (Zalles et al., 2019). This increased market in South America is also due to new advances in biological understanding and technologies following from them, as well as increasing investments by the major players in this market (<https://www.fortunebusinessinsights.com/industry-reports/biopesticides-market-100073>). This trend is likely to be seen in many other parts of the world in due course.

Finally, in Europe, the “Farm to Fork” strategy is a new challenge, designed to create sustainable food systems that will reduce dependency on pesticides and antimicrobials, reduce excess fertilization, improve animal welfare and reverse biodiversity loss. This is driving the crop protection market towards a higher use of biological control. The stated aim is to reduce, by 2030, the overall use and risk of chemical pesticides by 50% and the use of more hazardous pesticides by 50% (https://ec.europa.eu/food/sites/food/files/safety/docs/f2f_action-plan_2020_strategy-info_en.pdf) (Zalles et al., 2019).

There is also a need to consider sources of research and development funding in relation to public attitudes. One specific action in the strategy (https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en) is “investing in environmentally friendly technologies” and large R&D programmes where academia and industry join forces are indeed part of this agenda.

13 | FINAL REMARKS

Biological control of plant diseases with living organisms is challenging because the biology of at least three organisms has to be considered in a variable environment. As shown in Table 1,

BOX 1 Regulation of BCAs. Example of risk assessment from England*

Author: Matthew Everatt, Defra (Department for Environment, Food & Rural Affairs, UK Government) and Rosalind Noble, BSPP

See <https://secure.fera.defra.gov.uk/phiw/riskRegister/plant-health/non-native-biocontrol-agents.cfm>

Non-native microbial biological control agents (MBCAs) are covered by plant health if they are considered to be a quarantine plant health pest. Non-native BCAs/strains that are not ordinarily resident in the wild in the United Kingdom and not a regular visitor to the United Kingdom are covered by Defra.

The Chemical Regulations Division (CRD) of the UK Health and Safety Executive generally has responsibility for the use of pathogens as biological control agents, particularly if the pathogens are used in formulated products. Defra will, however, get involved for non-native pathogens (as described above) and may do the first risk assessment review.

For a classical biological control agent (released as a whole organism), the review has the following stages:

- Applicant submits risk assessment (see below)
- Internal review by Defra
- External review by other bodies, e.g., Natural England, Advisory Committee on Releases to the Environment, Devolved Administrations, etc.
- Plant Health Risk Group review (monthly meeting to discuss plant health issues)
- Public, stakeholder and European partner consultation
- May go back to the Plant Health Risk Group if there are any outstanding concerns
- Ministerial submission

The classical biological control agent may need to be approved by CRD in these cases.

For an augmentative biological control agent, the review has at least the following stages:

- Applicant submits risk assessment
- Internal review by Defra
- External review

The process is shorter, as the organism is not expected to become established in the environment and any impact will be limited and transient. However, if it could establish, then it will have to follow the review process as for a classical agent (if released as a whole organism). If it is a formulated product, the ultimate responsibility for approval will be with CRD. After the internal and external review, it would go to CRD for approval.

An application would look like this:

- Applicant and product information
- Background information on the MBCA
 - a. Identity
 - b. Distribution
 - c. Biology
- Voucher specimen/identity confirmation
 - a. This should be deposited in a recognized reference collection or the identity of the agent should be confirmed to prove the identity of the MBCA
- Risk assessment for the MBCA
 - a. Establishment
 - b. Host range
 - c. Spread
 - d. Direct/indirect impacts
 - e. Potential benefits
 - f. Uncertainty
 - g. Conclusion of the risk assessment
- Post release monitoring and control measures

For augmentative agents, the risk assessment portion may not need to cover host range, spread or direct/indirect impacts if the applicant is able to show that the agent is unable to establish in the United Kingdom.

*NB: Scotland and Wales have their own regulatory systems.

TABLE 2 Challenges and risks during product development of biological control agents (BCAs)

Stage	Challenge	Choices	Risk
Selection of isolate	Access and benefit sharing requirements re. sourcing and future use?	Choose best currently available isolates or search for better	Nagoya protocol on access and benefit sharing
Development	Production	Wet or dry fermentation	Cost effectiveness
	Formulation	Powder, liquid	
	Shelf life	Temperature and humidity during storage, formulation	Requirements too stringent (e.g., -20°C)
	Compatibility with existing technologies	Mix with other products	No suitable mixes
Delivery systems	Seed treatment (seed coating, biopriming, etc.)	Use of existing equipment	Specialist equipment needed
	Incorporation in growth substrates, spray application for upper part of plants	Growth substrate, incorporation method	Incompatible with biome in the medium
	Drench, broadcast, in furrow	Use of existing equipment or specialist development	
	Dusting, spraying, vector dispersal	As above	
Regulatory and industrial approval	Risk assessment (EU, EPA, etc.)	Scenarios	Refusal, or onerous conditions
	Field performance – GEP efficacy	Scale and scope of testing	Not quite good enough
	Ecology of the BCA and antagonist	A research-intensive part of the development	Unfavourable pathogen interactions
Full commercialization	Market size and market introduction	Partners, advisory support, publicity, pricing policy	Market too small to recoup development costs

Abbreviations: EPA, Environmental Protection Agency, USA; GEP, good experimental practice.

much progress has been made over recent decades, but much more development is needed for individual disease systems before these methods can be considered to be mature and a natural part of disease management technologies, as disease resistance and pesticides are today. At the biological level, scientific progress on understanding ecology and the biological (cellular/molecular) mechanisms governing the outcome of interactions alone and in combination is needed. By understanding these, there will be a rational basis for strain improvement, formulation and delivery, which will result in improved efficacy and stability. The political landscape, especially the green lobby, needs to be realistic about what can be achieved and the risks that need to be addressed. We are coming ever closer to being able to answer the question “what will it take to progress biologicals from ‘niche markets’ to broad acre crops and industrialized farming?” The pressures for reducing the use of pesticides in farming certainly provide an incentive to do this.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analysed.

ORCID

David B. Collinge  <https://orcid.org/0000-0001-6869-3585>

Dan Funck Jensen  <https://orcid.org/0000-0002-8855-5283>

Mojgan Rabiey  <https://orcid.org/0000-0001-8463-9087>

Sabrina Sarrocco  <https://orcid.org/0000-0002-7080-8369>

Michael W. Shaw  <https://orcid.org/0000-0002-6993-048X>

Richard H. Shaw  <https://orcid.org/0000-0003-0805-1841>

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